DUBLIN CITY UNIVERSITY

SEMESTER TWO REPEAT EXAMINATION 2018 – 2019

MODULE	**MS200 – Linear Algebra**
PROGRAMME(S)	B. Sc. in Computer Applications B. Sc. in Science Education B. Sc. in Physical Education with Mathematics B. Sc in Environmental Science and Health
YEAR OF STUDY	2
EXAMINER(S)	Prof. D. Wraith Dr. M. Clancy (Ext. 5774)
TIME ALLOWED	**2** hours
INSTRUCTIONS	**Answer ALL Questions.**
REQUIREMENTS	Log tables to be supplied. Approved calculators may be used.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO.
The use of programmable or text-storing calculators is expressly forbidden.

QUESTION 1

Throughout this question $\boldsymbol{u}$ and $\boldsymbol{v}$ will be the following vectors in $\mathbb{R}^3$:

$$\boldsymbol{u} = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} \quad \text{and} \quad \boldsymbol{v} = \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}.$$

Now, do the following:

(a) Calculate the vector $\boldsymbol{w} = 4\boldsymbol{v} - 2(\boldsymbol{u} + 2\boldsymbol{v})$.

[3 marks]

(b) Calculate the length of the vector $\boldsymbol{v}$.

[3 marks]

(c) If two non-zero vectors $\boldsymbol{x}$ and $\boldsymbol{y} \in \mathbb{R}^3$ are perpendicular, that is if $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$, show that $||\, \boldsymbol{x} + \boldsymbol{y} \,||^2 = ||\, \boldsymbol{x} \,||^2 + ||\, \boldsymbol{y} \,||^2$.

Hint: $||\, \boldsymbol{x} + \boldsymbol{y} \,||^2 = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{x} + \boldsymbol{y} \rangle$

[4 marks]

QUESTION 2

Throughout this question $\boldsymbol{u}_1$, $\boldsymbol{u}_2$ and $\boldsymbol{v}$ will be the following vectors in $\mathbb{R}^3$:

$$\boldsymbol{u}_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad \boldsymbol{u}_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \quad \text{and} \quad \boldsymbol{v} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}.$$

If $\boldsymbol{\mathcal{P}}$ is the plane through the origin for which $\{\boldsymbol{u}_1, \boldsymbol{u}_2\}$ is an **orthonormal** basis, do the following:

(a) Find the vector $\boldsymbol{v}^* \in \boldsymbol{\mathcal{P}}$ that is nearest to $\boldsymbol{v}$ and draw a diagram to illustrate $\boldsymbol{\mathcal{P}}$, $\boldsymbol{v}$ and $\boldsymbol{v}^*$.

[5 marks]

(b) Find a **non-zero** vector that is perpendicular to the plane $\boldsymbol{\mathcal{P}}$.

[5 marks]

QUESTION 3: Throughout this question $\boldsymbol{\mathcal{L}}$ will denote the line in $\mathbb{R}^3$ that passes through the point $\boldsymbol{p}$ and is in the direction of the vector $\boldsymbol{v}$, where

$$\boldsymbol{p} = \begin{bmatrix} 7 \\ 1 \\ 3 \end{bmatrix} \quad \text{and} \quad \boldsymbol{v} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}.$$

Do the following:

(a) Parametrize the line $\boldsymbol{\mathcal{L}}$.

[3 marks]

(b) Write down an equation for the plane that passes through the origin and intersects $\boldsymbol{\mathcal{L}}$ orthogonally.

[3 marks]

(c) Using parts (a) and (b), or otherwise, find (among all points on $\boldsymbol{\mathcal{L}}$) the point $\boldsymbol{q} \in \boldsymbol{\mathcal{L}}$ that is nearest to the origin.

[4 marks]

QUESTION 4: If the matrices A, B and $\boldsymbol{x}$ are given by:

$$A = \begin{bmatrix} 1 & -3 \\ 0 & -1 \\ 2 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & -2 & -1 \\ -1 & 4 & 1 \end{bmatrix}, \qquad \boldsymbol{x} = \begin{bmatrix} 1 \\ 1 \\ -5 \end{bmatrix},$$

do the following:

(a) State which (if any) of the following statements is/are true:

(i) The product $A\boldsymbol{x}$ is defined.

(ii) Both products AB and BA are defined.

(iii) Both products AB and BA are defined and $AB \neq BA$.

[6 marks]

(b) If an $(n \times n)$ matrix M satisfies the equation $M^2 + M - 6I = 0$, where I is the $(n \times n)$ identity - matrix, is it possible that $\lambda = 4$ is an eigenvalue of M? Justify your answer.

[4 marks]

QUESTION 5

(a) Write down the augmented matrix corresponding to the system of linear equations:

$$\boxed{\begin{array}{rcrcr} x & - & 7y & = & -1 \\ 3x & & & = & 2 \\ -x & + & 4y & = & 5 \\ 2x & + & y & = & 1 \end{array}}$$

[2 marks]

(b) For the remainder of this question the variables v, w, x, y, and z will satisfy a system of linear equations whose **augmented matrix** is $[\,A\,|\,\boldsymbol{b}\,]$. If the **reduced row echelon form** of $[\,A\,|\,\boldsymbol{b}\,]$ is:

$$\left[\begin{array}{ccccc|c} 1 & 2 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & -3 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array}\right],$$

do the following:

(i) Parametrize (as a subset of $\mathbb{R}^5$) the solution space of this system of equations in the variables v, w, x, y, and z.

[5 marks]

(ii) Determine a basis for the **kernel** of the matrix A and determine also the dimension of the **image** of A.

[3 marks]

QUESTION 6: Throughout this question the vector $\boldsymbol{u} \in \mathbb{R}^2$, the vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3 \in \mathbb{R}^3$ and the matrix A will be as follows:

$$\boldsymbol{u} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, \quad \boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}, \quad \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}, \quad \boldsymbol{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 2 & -1 \\ -2 & -4 & 2 \end{bmatrix}.$$

Answer the following, **making sure that in each case the logic of your argument is explained clearly**:

(a) Are the vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3 \in \mathbb{R}^3$ **linearly independent**?

[4 marks]

(b) What is the dimension of the **image** of A?

[3 marks]

(c) Is the vector $\boldsymbol{u}$ in the **image** of A?

[3 marks]

QUESTION 7

Throughout this question $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$ will denote the usual basis for $\mathbb{R}^3$. L_1 and L_2 will denote the rotations of $\mathbb{R}^3$ whose matrix representations relative to $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$ are M_{L_1} and M_{L_2}, respectively. If M_{L_1} and L_2 are specified by

$$\left\{ M_{L_1} = \frac{1}{2}\begin{bmatrix} 1 & \sqrt{2} & 1 \\ 1 & -\sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \end{bmatrix} \right\} \quad \text{and} \quad L_2 = \left\{ \begin{array}{c} \text{Anti-clockwise rotation} \\ \text{through } \pi/2 \text{ about} \\ \text{the } \boldsymbol{e}_1 \text{ - axis.} \end{array} \right\},$$

do the following:

(a) Calculate the matrix M_{L_2}.

[3 marks]

(b) Calculate the matrix $M_{L_1 \circ L_2}$ that represents the composition $L_1 \circ L_2$, relative to the usual basis for $\mathbb{R}^3$.

[3 marks]

(c) Find the axis of rotation of the composition $L_1 \circ L_2$.

[4 marks]

QUESTION 8

If the matrix $A = \begin{bmatrix} -4 & 0 & 4 \\ -3 & -3 & 9 \\ -2 & 0 & 2 \end{bmatrix}$, do the following:

(a) Find the **characteristic polynomial** of A and, hence, find all of the eigenvalues of A.

[5 marks]

(b) Parametrize the **eigenspace** of A that corresponds to the eigenvalue $\lambda = 0$. Is it possible to diagonalize A? Justify your answer.

[5 marks]

QUESTION 9

The 4×4 real matrix A has eigenvalues $\lambda = -3$ and $\lambda = 5$. Given that the corresponding eigenspaces are:

$$E_{-3} = \left\{ \begin{bmatrix} 2x - 3y + z \\ x \\ y \\ z \end{bmatrix} \mid x,\, y\, z \in \mathbb{R} \right\} \text{ and } E_5 = \left\{ \begin{bmatrix} 2z \\ 0 \\ -z \\ z \end{bmatrix} \mid z \in \mathbb{R} \right\},$$

respectively, do the following:

(a) Find a 4×4 invertible real matrix P and a 4×4 real **diagonal matrix** Λ such that

$$P^{-1}AP = \Lambda.$$

Note : You are **NOT** asked to calculate P^{-1}.

[6 marks]

(b) Determine, **with justification**, whether or not the equation $A\boldsymbol{x} = 2\boldsymbol{x}$ has a **non-zero** solution $\boldsymbol{x} \in \mathbb{R}^4$.

[4 marks]

QUESTION 10

(a) Write down the 4×4 elementary matrix E, such that the matrix EA is obtained from A when:

the **third** row of A is replaced by (row **three** of A - $2 \times$ row **two** of A).

[4 marks]

(b) Use Gauss-Jordan elimination to find the inverse of the matrix:

$$A = \begin{bmatrix} 1 & 0 & -1 \\ -2 & 1 & 4 \\ 0 & 1 & 1 \end{bmatrix}$$

[6 marks]